Statistics 6910:

Case: Behavioral Risk Factor Surveillance System (BRFSS) Survey

BRFSS Survey

- Centers for Disease Control and Prevention (CDC) developed a survey with a standard core questionnaire to provide data to the states on health
- World's largest, on-going telephone health survey system
- Tracks health conditions and risk behaviors in the United States

Information available for each subject

- Weight (pounds)
- Desired weight (pounds)
- Height (inches)
- Gender
- Age (years)
- …

Body Mass Index (BMI)

- Measure of obesity
- Ratio: weight in kilograms to height^2 in meters
- 2.54 cm to the inch
- 2.2 pounds to the kilogram
- TO DO: Write an equation for computing BMI for someone with weight measured in pounds and height measured in inches

Body Mass Index

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$$BMI = \frac{weight / 2.2}{\left(\frac{2.54}{100} height\right)^2}$$

BMI = (weight / 2.2) / (2.54 / 100 * height)^2
Note that () are optional and can be omitted.

Let's Examine the Data in R

What have we observed?

- R works with variables, which are called vectors
- R's computations are vectorized
- R has values for Not Available (NA) and Infinity (Inf)
- When we calculate averages, if one value is NA then the average is NA, unless we tell the function to ignore the NAs
- Aside from numeric data, R has factor data, which is especially designed for qualitative (categorical) information.

Let's formalize our observations